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MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	CSM-B10

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Title of the Project: Enhanced AA-CLIP for Robust Zero-Shot Visual Anomaly Detection and Localization

Abstract

Visual anomaly detection is an important problem in automated inspection, where quality-control personnel need to identify defective or abnormal regions in product images. Traditional anomaly detection methods often require class-specific training data, making it difficult to detect previously unseen objects and defect types. Vision-language models such as CLIP provide strong generalization capabilities, but the original CLIP representation is not specifically optimized to distinguish between normal and anomalous visual features, limiting its effectiveness for zero-shot anomaly detection. Existing research, including AnomalyCLIP and AA-CLIP, addresses this limitation by adapting CLIP for anomaly detection. The selected base paper, AA-CLIP: Enhancing Zero-Shot Anomaly Detection via Anomaly-Aware CLIP, introduces anomaly-aware text representations, residual adapters, and patch-level visual feature alignment to improve anomaly detection and localization. However, challenges still remain in accurately identifying small or subtle anomalies, maintaining robustness under different image conditions, and achieving reliable generalization across different objects and domains. Therefore, this research will build upon AA-CLIP and investigate an enhanced approach focusing on fine-grained anomaly localization, robustness, and prediction reliability. The proposed work will first establish the original AA-CLIP as a baseline and then experimentally investigate strategies for improving the representation and localization of small anomalous regions. The proposed approach will be compared with CLIP and AA-CLIP using image-level and pixel-level evaluation metrics such as AUROC, on established anomaly detection datasets including MVTec AD, VisA, BTAD, and MPDD. The project will focus on research and experimental evaluation of zero-shot visual anomaly detection and localization and will not include deployment in safety-critical industrial environments. The expected outcome is an experimentally validated enhancement of AA-CLIP that provides more accurate anomaly localization and improved robustness and reliability while retaining the zero-shot generalization capability of the base approach.

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References (Category specific)

1. ***Ma, W., Zhang, X., Yao, Q., Tang, F., Wu, C., Li, Y., Yan, R., Jiang, Z., & Zhou, S. K. (2025)***. *AA-CLIP: Enhancing Zero-Shot Anomaly Detection via Anomaly-Aware CLIP. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025).* ***[Base Paper]***

2. ***Zhou, Q., Pang, G., Tian, Y., He, S., & Chen, J. (2024).*** *AnomalyCLIP: Object-Agnostic Prompt Learning for Zero-Shot Anomaly Detection. International Conference on Learning Representations (ICLR 2024).*

3. ***Radford, A., Kim, J. W., Hallacy, C., et al. (2021).*** *Learning Transferable Visual Models From Natural Language Supervision. Proceedings of the 38th International Conference on Machine Learning (ICML).*

4. ***Roth, K., Pemula, L., Zepeda, J., Schölkopf, B., Brox, T., & Gehler, P. (2022).*** *Towards Total Recall in Industrial Anomaly Detection. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022).*

5. ***Zou, Y., Jeong, J., Pemula, L., Zhang, D., & Dabeer, O. (2022).*** *SPot-the-Difference Self-Supervised Pre-training for Anomaly Detection and Segmentation. European Conference on Computer Vision (ECCV 2022), 392–408.*

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